CS182: Introduction to Machine Learning
Lecture 03: Probability and Statistics

Worked Solutions and Marking Guide

Scope: Probability_Statistics.pdf. Independent practice material based on the official 2026 Fall lecture.

Total: 100 marks. **Suggested time:** 120 minutes.

Section	Questions	Each	Marks
A. Misconceptions: True/False	12	1	12
B. Multiple response	12	2	24
C. Proofs and formula derivations	4	10	40
D. Applications	2	12	24
Total	30		100

Scoring. In Section B each item has exactly four options, of which one, two, or three are correct. The complete correct set earns 2 marks; a nonempty proper subset earns 1 mark; any incorrect option or no selection earns 0. T/F answers earn 1 mark only when correct. Show reasoning in Sections C and D; part marks are available for valid intermediate work. Equivalent correct methods receive credit.

Instructions. Answer in English. Use natural logarithms. Exact values and justified approximations are accepted. All vectors are columns unless stated otherwise. Use local assumptions, label conventions, and tie rules.

Quick key — Section A

1	2	3	4	5	6
F	F	F	F	F	F

7	8	9	10	11	12
T	T	T	T	T	F

Quick key — Section B

13	14	15	16	17	18
BD	D	AC	A	BD	ACD

19	20	21	22	23	24
ABC	BC	AD	BD	AC	AD

Section A. True/False — Explanations

1. False. Mutual exclusivity gives $P(A \cap B) = 0$, whereas independence would require $P(A \cap B) = P(A)P(B) > 0$.
Source: PDF pp.4-7.

2. False. The central limit theorem gives a limiting distribution, hence a large-sample approximation. Exact finite-sample normality follows when the observations themselves are iid normal.
Source: PDF pp.29-30,37.

3. False. The slide gives $\mathrm{Var}(T) = \nu/(\nu - 2)$ for $\nu > 2$. Thus t_3 has variance 3, reflecting uncertainty in the random denominator.
Source: PDF pp.35-36.

4. False. The sufficient condition is joint Gaussianity. Let $Z \sim N(0, 1)$ and let an independent sign R equal ± 1 with equal probabilities. Then $X = Z$ and $Y = RZ$ each are standard normal and have covariance zero, but $|X| = |Y|$, so they are dependent. This states the qualification needed for the normal-variable claim on the slide.
Source: PDF pp.20,29,38.

5. False. Linearity of expectation needs existence of the relevant expectations, not independence. Independence is relevant to factoring product expectations and eliminating covariance terms.
Source: PDF pp.18-20.

6. False. The law concerns convergence in probability: for each fixed $\epsilon > 0$, $P(|\bar{X}_n - \mu| > \epsilon) \to 0$. It does not assert exact equality at a finite sample size.
Source: PDF p.23.

7. True. The mean has weights $1/n$; independence eliminates all off-diagonal covariance terms, giving $n\sigma^2/n^2$.
Source: PDF pp.20,23.

8. True. A density is integrated to obtain probabilities. A uniform variable on $(0, 1/4)$ has density 4 there, and every singleton has probability zero.
Source: PDF pp.12,28.

9. True. Expand the centered difference and take expectations; the cross term has a minus sign.
Source: PDF p.20.

10. True. Centering removes one orthogonal Gaussian coordinate, leaving the sum of squares of $n - 1$ independent standard normals. The factor is $(n-1)S^2/\sigma^2$, which also corrects the scaling typographical error on p.37.
Source: PDF pp.31-34,37.

11. True. This is the construction of χ^2_k; its mean is k and its variance is $2k$. Independence and unit normal scaling both matter.
Source: PDF p.31.

12. False. Bayes' rule gives $P(A \mid B)/P(B \mid A) = P(A)/P(B)$ when the common joint probability is positive. For example, $B \subset A$ can make the two conditional probabilities different.
Source: PDF pp.5-8.

Section B. Multiple Response — Explanations

13. Correct set: B, D.

(A) **Incorrect.** The assumptions do not say the observations are normal; the central limit theorem supplies an asymptotic approximation.

(B) **Correct.** This is the weak law of large numbers under the supplied assumptions.

(C) **Incorrect.** The variance of the sum is $9n$, so dividing by n^2 gives $9/n$.

(D) **Correct.** The central limit theorem approximates the centered and scaled mean by a standard normal.

Source: PDF pp.20,23,30.

14. Correct set: D.

(A) **Incorrect.** The final factor must condition on both earlier events: $P(C \mid A \cap B)$. Conditioning only on B needs an additional conditional-independence assumption.

(B) **Incorrect.** The denominator is the probability of the conditioning event B.

(C) **Incorrect.** The conditional terms must be weighted by $P(A_1)$ and $P(A_2)$.

(D) **Correct.** The intersection is counted twice by the first two terms and must be subtracted once.

Source: PDF pp.3-8.

15. Correct set: A, C.

(A) **Correct.** At a fixed x, integrate 2 over $x < y < 1$.

(B) **Incorrect.** The triangular support is not the product of the marginal supports. The joint density does not equal $p_X(x)p_Y(y)$.

(C) **Correct.** The triangular support has area $1/2$, so multiplying by two gives total mass one.

(D) **Incorrect.** The marginal is $p_Y(y) = 2y$, giving conditional density $1/y$ on $(0, y)$.

Source: PDF pp.12,15-17.

16. Correct set: A.

(A) **Correct.** Compute the finite sums over the three equally likely values.

(B) **Incorrect.** Knowing $X = 0$ forces $Y = 0$; $P(Y = 0 \mid X = 0) = 1 \neq 1/3$.

(C) **Incorrect.** Correlation requires positive variances, not nonzero means. Here $\mathrm{Var}(X) = 2/3$, $\mathrm{Var}(Y) = 2/9$, and the correlation is zero.

(D) **Incorrect.** Here $E[XY] = E[X^3] = 0$ and $E[X] = 0$, so the covariance is zero. The number $2/9$ is the variance of Y, not this covariance.

Source: PDF pp.18-20.

Section B. Multiple Response — Explanations

17. Correct set: B, D.

(A) **Incorrect.** Each marginal count is binomial with ten trials; a Bernoulli variable only takes values zero and one.

(B) **Correct.** Each trial contributes to exactly one category count.

(C) **Incorrect.** The fixed total couples the category counts. For example, $N_1 = 10$ forces $N_2 = 0$.

(D) **Correct.** The multinomial coefficient counts distinct arrangements of the category outcomes.

Source: PDF pp.25-27.

18. Correct set: A, C, D.

(A) **Correct.** A constant shift vanishes after centering; the scale is squared.

(B) **Incorrect.** The missing term is $\mathrm{Var}(X)$, which need not be zero.

(C) **Correct.** Centering removes constants and multiplication contributes the product $2(-1)$.

(D) **Correct.** Expand the centered square and use linearity.

Source: PDF pp.18-20.

19. Correct set: A, B, C.

(A) **Correct.** They are jointly Gaussian with zero covariance, so the joint density factors.

(B) **Correct.** The inverse covariance is diagonal with entries $1/4$ and $1/9$.

(C) **Correct.** The determinant is 36, whose positive square root is six.

(D) **Incorrect.** The density at the mean is $1/(12\pi)$, while the probability of that exact point is zero.

Source: PDF pp.12,29,38.

20. Correct set: B, C.

(A) **Incorrect.** Variance scales by the square of the divisor, so the variance is $2.55/400$.

(B) **Correct.** There are $\binom{20}{2}$ disjoint placements of the two successes, each with the displayed product probability.

(C) **Correct.** Add the expectations of the twenty success indicators.

(D) **Incorrect.** The variance is $20(0.15)(0.85) = 2.55$.

Source: PDF pp.20,25-26.

Section B. Multiple Response — Explanations

21. Correct set: A, D.

(A) **Correct.** They depend on disjoint coordinates after an orthogonal transformation of the standardized Gaussian sample.

(B) **Incorrect.** The residual sum of squares has $n-1$ degrees of freedom, so the ratio has distribution t_{n-1}.

(C) **Incorrect.** The correct residual sum is $(n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$.

(D) **Correct.** A normalized linear combination of independent normal variables is exactly standard normal.

Source: PDF pp.31-35,37.

22. Correct set: B, D.

(A) **Incorrect.** The t distribution approaches the standard normal as degrees of freedom grow; finite degrees of freedom retain heavier tails.

(B) **Correct.** This is the location and scale transformation in the general t density.

(C) **Incorrect.** The scale parameter squared is four, but $\mathrm{Var}(T) = 5/3$, so $\mathrm{Var}(W) = 20/3$.

(D) **Correct.** The mean of t_5 exists and is zero.

Source: PDF pp.35-36.

23. Correct set: A, C.

(A) **Correct.** Conditioning on both bits being one forces them to be equal.

(B) **Incorrect.** This construction is pairwise independent, but its triple intersection fails the product rule for mutual independence.

(C) **Correct.** Both have probability $1/2$ and their intersection has probability $1/4$.

(D) **Incorrect.** Their triple intersection has probability $1/4$, whereas the product of the three probabilities is $1/8$.

Source: PDF pp.5-7.

24. Correct set: A, D.

(A) **Correct.** Sum the joint probabilities across both values of Y: $0.20 + 0.40 = 0.60$.

(B) **Incorrect.** The conditional probability is $0.30/(0.30 + 0.20) = 0.60$.

(C) **Incorrect.** The product of marginals at $(1, 1)$ is $0.60 \cdot 0.50 = 0.30$, not 0.40.

(D) **Correct.** Divide 0.40 by the marginal 0.60.

Source: PDF pp.13-17.

Section C. Proofs and Formula Derivations

25. Deriving total probability, Bayes' rule, and posterior odds [10 marks]

(a) [3 marks] The events $B \cap A_i$ are disjoint and their union is B. Therefore

$$P(B) = \sum_{i=1}^{k} P(B \cap A_i) = \sum_{i=1}^{k} P(B \mid A_i)P(A_i).$$

To condition on B, require this sum, and hence $P(B)$, to be positive.

(b) [3 marks] By conditional probability and the product rule,

$$P(A_i \mid B) = \frac{P(A_i \cap B)}{P(B)} = \frac{P(B \mid A_i)P(A_i)}{\sum_j P(B \mid A_j)P(A_j)}.$$

Summing the numerators over i reproduces the common positive denominator, so the posteriors sum to one.

(c) [4 marks] Cancel the common evidence denominator in Bayes' rule to obtain

$$\frac{P(H_1 \mid E \cap F)}{P(H_0 \mid E \cap F)} = \frac{P(H_1)}{P(H_0)} \frac{P(E \cap F \mid H_1)}{P(E \cap F \mid H_0)}.$$

The conditional product rule gives $P(E \cap F \mid H_i) = P(E \mid H_i)P(F \mid E, H_i)$. If E and F are conditionally independent given each hypothesis, this becomes $P(E \mid H_i)P(F \mid H_i)$. Thus posterior odds equal prior odds times two likelihood ratios. Unconditional independence of the two observations is not a substitute for this assumption.

Marking: use the displayed subpart allocations; award credit for the logical or numerical intermediate steps.

Source: PDF pp.5-8.

Section C. Proofs and Formula Derivations

26. Variance formulas and the effect of shared noise **[10 marks]**

(a) [3 marks] Writing $\mu_X = E[X]$ and $\mu_Y = E[Y]$, expansion yields $E[(X - \mu_X)(Y - \mu_Y)] = E[XY] - \mu_X\mu_Y$. Also $aX + bY + c - E[aX + bY + c] = a(X - \mu_X) + b(Y - \mu_Y)$. Squaring and taking expectations gives

$$\mathrm{Var}(aX + bY + c) = a^2 \mathrm{Var}(X) + b^2 \mathrm{Var}(Y) + 2ab \,\mathrm{Cov}(X, Y).$$

(b) [4 marks] Linearity gives $E[\bar{X}_n] = \mu$. Expanding the centered sum gives

$$\mathrm{Var}(\bar{X}_n) = \frac{1}{n^2} E\left[\sum_i (X_i - \mu) \sum_j (X_j - \mu)\right] = \frac{1}{n^2} \sum_{i,j} \mathrm{Cov}(X_i, X_j).$$

Equivalently it is $\sigma^2/n + (2/n^2) \sum_{i<j} \mathrm{Cov}(X_i, X_j)$. Under independence, each off-diagonal term is zero, so $\mathrm{Var}(\bar{X}_n) = \sigma^2/n$. Identical means alone do not eliminate these terms.

(c) [3 marks] We have $\bar{X}_n = \mu + B + n^{-1} \sum_i \epsilon_i$. Independence gives

$$\mathrm{Var}(\bar{X}_n) = \tau^2 + \frac{\sigma_\epsilon^2}{n}.$$

Alternatively, each individual variance is $\tau^2 + \sigma_\epsilon^2$, and every distinct pair has covariance τ^2, which gives the same result in the formula above. The measurements share B and are not independent. Averaging removes the independent noise component but leaves the positive variance floor τ^2.

Marking: use the displayed subpart allocations; award credit for the logical or numerical intermediate steps.

Source: PDF pp.18-20,23.

Section C. Proofs and Formula Derivations

27. Constructing the binomial distribution from trials [10 marks]

(a) [3 marks] A fixed ordered sequence containing k successes and $n-k$ failures has probability $p^k(1-p)^{n-k}$ by independence. Choosing the k success positions gives $\binom{n}{k}$ disjoint sequences. Therefore

$$P(K=k)=\binom{n}{k}p^k(1-p)^{n-k}, \qquad k=0,\dots,n.$$

(b) [3 marks] For an indicator I, $E[I]=0(1-p)+1p=p$, and $I^2=I$, so $\mathrm{Var}(I)=E[I^2]-(E[I])^2=p-p^2=p(1-p)$. Linearity then gives $E[K]=\sum_i E[I_i]=np$; this expectation calculation alone does not need independence.

(c) [4 marks] Expand the centered square:

$$\mathrm{Var}(K)=\sum_i \mathrm{Var}(I_i)+2\sum_{i<j}\mathrm{Cov}(I_i,I_j).$$

Independence implies $E[I_iI_j]=p^2$, so the cross covariances vanish and $\mathrm{Var}(K)=np(1-p)$. Scaling by $1/n$ gives $\mathrm{Var}(K/n)=p(1-p)/n$. With dependent indicators the general expression is $np(1-p)+2\sum_{i<j}\mathrm{Cov}(I_i,I_j)$; the binomial pmf is also no longer guaranteed.

Marking: use the displayed subpart allocations; award credit for the logical or numerical intermediate steps.

Source: PDF pp.5-7,18-20,25-26.

Section C. Proofs and Formula Derivations

28. An exact Gaussian mean and residual decomposition [10 marks]

(a) [3 marks] The independent standardized observations have joint density $(2\pi)^{-n/2}\exp(-z^Tz/2)$. Since $z = A^Ty$, $z^Tz = y^Ty$ and the absolute Jacobian determinant is $|\det A^T| = 1$. Thus

$$p_Y(y) = (2\pi)^{-n/2}e^{-\sum_j y_j^2/2} = \prod_{j=1}^{n}\frac{1}{\sqrt{2\pi}}e^{-y_j^2/2}.$$

This factorization proves that the coordinates are iid standard normal. The first row has norm one because its entries are $1/\sqrt{n}$.

(b) [4 marks] The first coordinate is $Y_1 = n^{-1/2}\sum_i Z_i = \sqrt{n}(\bar{X}-\mu)/\sigma$. The residual identity is

$$\frac{\sum_i(X_i-\bar{X})^2}{\sigma^2} = \sum_i(Z_i-\bar{Z})^2 = \sum_i Z_i^2 - n\bar{Z}^2 = \sum_{j=2}^{n}Y_j^2.$$

The last sum is χ^2_{n-1}, since it contains $n-1$ independent squared standard normals. It is a function only of $Y_2,\dots,Y_n$, and hence is independent of Y_1. The sample mean is a function of Y_1; S^2 is a function of the remaining coordinates. They are independent.

(c) [3 marks] Let $U = (n-1)S^2/\sigma^2 = \sum_{j=2}^{n}Y_j^2$. Then $U \sim \chi^2_{n-1}$ and U is independent of $Y_1 \sim N(0,1)$. Therefore

$$\frac{\sqrt{n}(\bar{X}-\mu)}{S} = \frac{Y_1}{\sqrt{U/(n-1)}} \sim t_{n-1}.$$

For general iid finite-variance data, the normal-coordinate density factorization and exact chi-square residual law need not hold. The central limit theorem gives asymptotic normal behavior for a standardized mean, not this finite-sample t law.

Marking: use the displayed subpart allocations; award credit for the logical or numerical intermediate steps.

Source: PDF pp.29,31-35,37-38.

Section D. Applications

29. Two inspections and the base rate of defective components [12 marks]

(a) [4 marks] Let D denote a defect and F a flag. Total probability gives $P(F) = 0.90(0.03) + 0.04(0.97) = 0.0658$. Bayes' rule gives

$$P(D \mid F) = \frac{0.027}{0.0658} = \frac{135}{329} \approx 0.410334.$$

Even with high sensitivity, nondefective components are common enough to contribute substantial false flags.

(b) [4 marks] Conditional independence gives $P(F_1 \cap F_2 \mid D) = 0.90^2$ and $P(F_1 \cap F_2 \mid D^c) = 0.04^2$. Thus

$$P(F_1 \cap F_2) = 0.03(0.81) + 0.97(0.0016) = 0.025852,$$

and

$$P(D \mid F_1 \cap F_2) = \frac{0.0243}{0.025852} = \frac{6075}{6463} \approx 0.939966.$$

We factor within each defect class; the two flag events need not be independent without conditioning.

(c) [4 marks] Each flag indicator is Bernoulli with probability $q = 0.0658$, and the stated assumptions make the indicators independent. Hence $K \sim \mathrm{Bin}(200, 0.0658)$, $E[K] = 200q = 13.16$, and

$$\mathrm{Var}(K) = 200q(1-q) = 12.294072.$$

The fraction has variance $12.294072/200^2 = 0.0003073518$. Its mean is q, not the underlying defect probability 0.03.

Marking: use the displayed subpart allocations; award credit for the logical or numerical intermediate steps.

Source: PDF pp.5-8,18-20,25-26.

Section D. Applications

30. Exact inference for a normally distributed calibration error [12 marks]

(a) [4 marks] The deviations from 500 are $-2, 2, 1, -1, 0, 3, -3, 0, 0$, which sum to zero. Thus $\bar{x} = 500$ and the squared deviations sum to $4 + 4 + 1 + 1 + 0 + 9 + 9 = 28$. Therefore $s^2 = 28/8 = 3.5$ square millimeters, and $s/\sqrt{9} = \sqrt{3.5}/3 \approx 0.623610$ millimeters.

(b) [4 marks] The observed statistic is $t = (500 - 499)/(\sqrt{3.5}/3) = 3/\sqrt{3.5} \approx 1.603567$. Under the normal model, $T = \sqrt{9}(\bar{X} - \mu)/S \sim t_8$. The probability statement $P(-2.306 \le T \le 2.306) \approx 0.95$, with rounding in the quantile, rearranges to

$$\bar{X} - 2.306\frac{S}{3} \le \mu \le \bar{X} + 2.306\frac{S}{3}.$$

Substitution yields 500 ± 1.438044, or approximately $[498.561956, 501.438044]$ millimeters. Exact t coverage requires iid normal observations; finite variance by itself is insufficient.

(c) [4 marks] The observed chi-square statistic is $(9 - 1)s^2/\sigma_0^2 = 28/4 = 7$. For the unknown true variance, $U = 8S^2/\sigma^2 \sim \chi_8^2$. Inverting $2.180 \le 8S^2/\sigma^2 \le 17.535$ reverses the corresponding reciprocal inequalities and gives

$$\frac{8S^2}{17.535} \le \sigma^2 \le \frac{8S^2}{2.180}.$$

The interval is $[28/17.535, 28/2.180] \approx [1.596806, 12.844037]$ square millimeters. The coverage is approximately 95% because the supplied quantiles are rounded.

Marking: use the displayed subpart allocations; award credit for the logical or numerical intermediate steps.

Source: PDF pp.31-37; interval construction is a supplemental application of the lecture pivots.

Lecture Alignment and Coverage

Primary source: `Probability_Statistics.pdf` in `E:\CS182\2026fall\lecture`. All PDF pages count the cover as page 1. Additional arguments are marked as supplementary derivations when appropriate. These are original practice questions, not copied homework questions.

T/F

1 PDF pp.4-7.

2 PDF pp.29-30,37.

3 PDF pp.35-36.

4 PDF pp.20,29,38.

5 PDF pp.18-20.

6 PDF p.23.

7 PDF pp.20,23.

8 PDF pp.12,28.

9 PDF p.20.

10 PDF pp.31-34,37.

11 PDF p.31.

12 PDF pp.5-8.

Multiple response

13 PDF pp.20,23,30.

14 PDF pp.3-8.

15 PDF pp.12,15-17.

16 PDF pp.18-20.

17 PDF pp.25-27.

18 PDF pp.18-20.

19 PDF pp.12,29,38.

20 PDF pp.20,25-26.

21 PDF pp.31-35,37.

22 PDF pp.35-36.

23 PDF pp.5-7.

24 PDF pp.13-17.

Proofs and applications

25 PDF pp.5-8.

26 PDF pp.18-20,23.

27 PDF pp.5-7,18-20,25-26.

28 PDF pp.29,31-35,37-38.

29 PDF pp.5-8,18-20,25-26.

30 PDF pp.31-37; interval construction is a supplemental application of the lecture pivots.